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FORKLIFT

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QUALIFICATION TRAINING PACKAGE

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Chapter 1

OVERVIEW

1.1. Overview. This Qualification Training Package (QTP) establishes standardized training requirements, content, and certification protocols for all Department of the Air Force (DAF) operators of this vehicle. Designed to ensure consistency and excellence across all installations, this package provides comprehensive guidance for trainers, trainees, and certifiers throughout the certification process.

1.1.1. Instructional Standards. The training standards outlined herein align with DAF requirements and serve as the authoritative source for curriculum development and performance evaluation. Trainers will utilize **Chapters 2** through **Chapter 6** to guide trainee preparation, conduct classroom lectures, provide demonstrations, and administer knowledge and performance evaluations.

1.2. Applicability. This QTP applies to all DAF personnel tasked with operating the vehicles applicable to this package.

1.3. Administrative. Send comments and suggested improvements on DAF Form 847, *Recommendation for Change of Publication*, through Air Force Installation and Mission Support Center (AFIMSC) functional managers via e-mail at: AFIMSC.A34OG.GroundTrans@us.af.mil.

Chapter 2

RESPONSIBILITIES

2.1. Trainer. The trainer is responsible for the overall execution of the training plan. This includes providing an overview of **Chapters 2** and **3**, delivering classroom lectures based on **Chapter 4**, and conducting knowledge training and demonstrations (**Chapter 5**). The trainer will administer and grade all review questions, ensuring that any missed items are thoroughly reviewed with the trainee to guarantee task comprehension before signing off on the training block.

2.2. Trainee. The trainee must actively participate in the QTP process and meet all preparation requirements before beginning practical instruction.

2.2.1. Administrative Requirements: The trainee must possess a valid state driver's license and ensure Sections I and II of the DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Driver's License,* are completed in accordance with (IAW) Department of the Air Force Instruction (DAFI) 24-301, *Ground Transportation*. Additionally, all training and licensing must comply with any applicable local licensing jurisdiction requirements.

2.2.2. Mandatory Reading: The trainee must read this entire QTP and the manufacturer's operator's manual for the specific forklift being trained on.

2.2.3. Execution: During instruction, the trainee will follow along with the lecture, utilize **Attachments 2** and **4** as inspection guides, accomplish the forklift operation review questions **(Attachment 5)** with the trainer, and prepare for the unassisted performance test.

2.3. Certifier. The certifier acts as the independent evaluating authority. They are responsible for evaluating the Airman's physical task performance (**Chapter 6**) without providing any assistance or coaching. Upon successful, independent completion of the performance test, the certifier will add their name, date, and signature, to sign off on the task(s) finalizing the licensing process.

2.3.1. Recertification Requirement: Forklift operators must successfully complete the Performance Test and review questions every three years to maintain active certification. This triennial recertification mandate applies strictly to Powered Industrial Trucks and must be executed in accordance with all applicable local licensing jurisdiction requirements.

Chapter 3

INTRODUCTION

3.1. Objectives. The objective of this course is to ensure personnel become fully qualified forklift operators who possess the requisite knowledge and skills for safe, professional execution IAW Department of Defense Instruction 6055.04, *DoD Motor Vehicle and Traffic Safety*. Through a comprehensive curriculum of lectures, demonstrations, and hands-on driving sessions, trainees will be trained in both the safe operation and preventive maintenance of various forklift platforms. Ultimately, trainees must demonstrate their proficiency by independently performing operator inspections and completing the final performance test (Go/No-Go checklist) with zero instructor assists.

3.2. Desired Learning Outcomes.

3.2.1. Understand the safety precautions to be followed before-, during-, and after-operation of the forklift.

3.2.2. Understand the purpose of the forklift and its role in the mission.

3.2.3. Understand Occupational Safety & Health Administration (OSHA) and Air Force Occupational Safety and Health (AFOSH) compliance requirements.

3.2.4. Know the proper operator maintenance procedures of the forklift IAW applicable technical orders (TO) and use of DAF Form 1800, *Operator's Inspection Guide and Trouble Report*.

3.2.5. Safely and proficiently operate the forklift.

3.3. Lesson Duration. Recommended instructional and hands-on training time is 40 hours.

Table 3.1. Recommended Training Time for Training Activities. Note: The durations listed in Table 3.1 represent recommended baseline training times and are not absolute mandates. Actual training durations may be adjusted at instructor discretion to be shorter or longer, depending entirely on the trainee's progress and demonstrated task proficiency.

Training Activity	Training Time
Trainee's Preparation	2 Hours
Instructor's Lecture	2 Hours
Trainee's Written Evaluation	1 Hour
Instructor's Demonstration	4 Hours
Trainee's Personal Experience (to build confidence and proficiency) • Perform Operator Maintenance • Operate the Vehicle	30 Hours
Trainee's Performance Evaluation	1 Hour

3.4. Instructional References.

3.4.1. Risk Management (RM) and Safety Principles.

3.4.2. Applicable TOs or Manufacturer's Operator's Manual (see vehicle maintenance for TO number for vehicle being used in training).

3.4.3. DAF Form 1800.

3.4.4. DAFMAN 91-201, Explosive Safety Standards.

3.4.5. Regulations (Standards - 29 Code of Federal Regulations (CFR)), Part 1910 - Occupational Safety and Health Standards, Subpart N – Material Handling and Storage.

3.4.6. OSHA Directive, Enforcement and Compliance (CPL) 02-01-028 - CPL 2-1.28A - Compliance Assistance for the Powered Industrial Truck Operator Training Standards.

3.5. Instructional Training Aids and Equipment.

3.5.1. Forklift QTP.

3.5.2. Forklift.

3.5.3. Applicable TO or manufacturer's operator's manual.

3.5.4. DAF Form 1800.

3.5.5. Videos (if locally produced).

3.5.6. Suitable training area.

3.5.7. Traffic cones or suitable markers.

3.5.8. Palletized loads or skids.

3.5.9. Tractor/Trailer combination and/or loading dock.

3.5.10. K-Loader, rollerized trailer, or aircraft (for aircraft loading).

3.6. Suggested Sources for Obtaining Supplemental Training Material.

3.6.1. J.J. Keller & Associates - www.jjkeller.com

3.6.2. The Training Network - www.safetytrainingnetwork.com/products/forklift.shtml

Chapter 4

KNOWLEDGE AND OPERATOR PERFORMANCE

4.1. Forklift Design and Specifications. The Air Force uses a variety of forklifts (also referred to as "powered industrial trucks") ranging from simple hand lifts to complex powered, all-terrain machines. Forklifts vary significantly in size, shape, and specifications. It is imperative that operators know the specifications, rated loads, and limitations of the specific forklift they are operating.

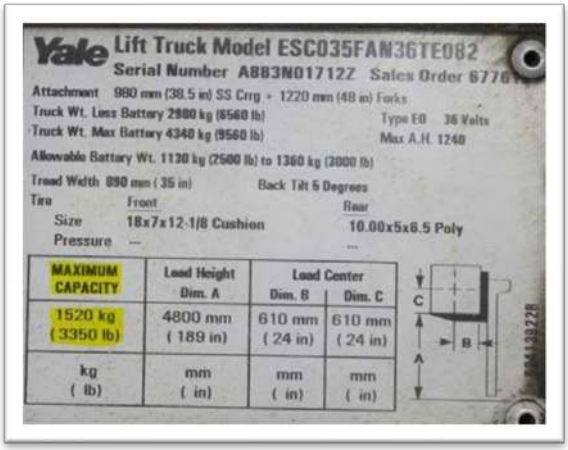
4.1.1. Data Plate. Specifications and rated load information must be used together to determine the proper use and operating environment for the forklift. This information is found in the appropriate Technical Order (TO) or directly on the vehicle data plate. (See Figure 4.1 for a forklift data plate example).

4.1.1.1. Specifications: Describe equipment dimensions, weight, engine/drivetrain data, turning radius, and fluid types.

4.1.1.2. Rated Load/Capabilities: Describe lifting capacity, load center, maximum reach, and maximum stacking height.

4.1.1.3. Warning: Specifications given are for a centered and balanced load. Lifting the maximum rated capacity when a load is unbalanced can cause the forklift to tip over.

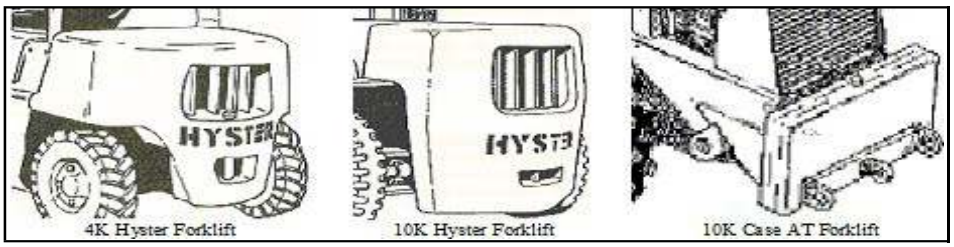
Figure 4.1. Forklift Data Plate Example.



4.2. Major Forklift Components. While the makeup of a forklift differs by make and model, all standard forklifts are equipped with three primary structural components: a counterweight, a lifting mast/device, and an overhead guard.

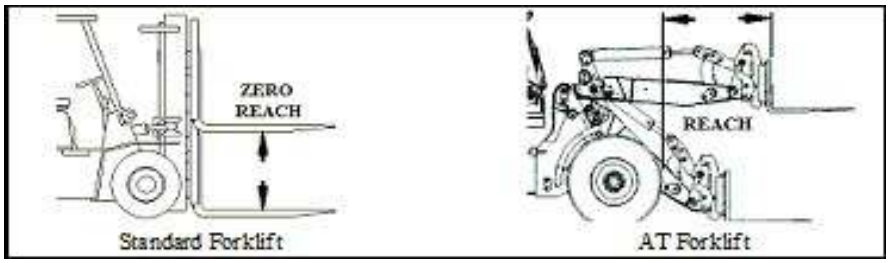
4.2.1. Counterweights. The counterweight opposes the downward force of the load placed on the forks, keeping the vehicle grounded and stable. Counterweights come in many shapes and sizes; some are removable or stackable to reduce weight when operating over soft terrain or for shipping. Because most counterweights are simply rested into place with minimal hardware, operators must verify they are securely fastened prior to operation. (See **Figure 4.2**).

Figure 4.2. Forklift Counterweight Styles.



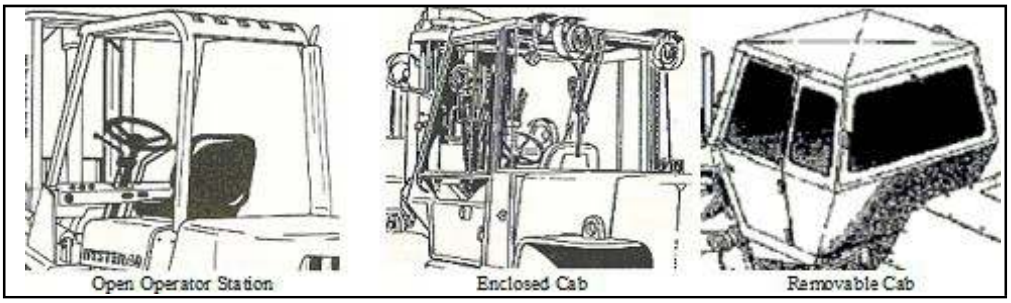
4.2.2. Lifting Mast or Device. Lifting devices vary heavily by make and function. Standard forklifts lift loads straight up and down, while all-terrain (AT) and telehandler models lift their masts at an angle and can "reach" outward. Some forklifts are equipped to rotate loads, while others have masts that swing laterally. Operators must familiarize themselves with the specific reach and mast style of their assigned equipment. (See Figure 4.3).

Figure 4.3. Lifting Device Reach.



4.2.3. Overhead Guards. Overhead guards protect the operator from falling objects and provide a crush-proof cage in the event of a tip-over. On forklifts with enclosed cabs, the overhead guard is built directly into the cab frame, offering additional protection from environmental elements. If a cab or guard is removed for shipping purposes, it must be completely reinstalled before operation resumes. (See Figure 4.4).

Figure 4.4. Overhead Guards.



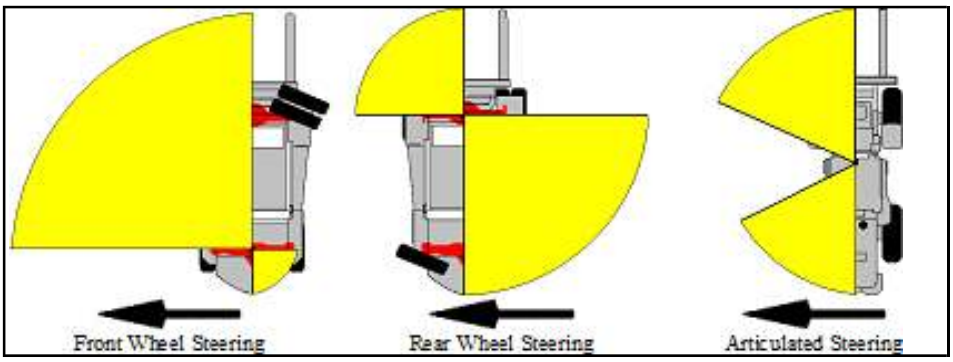
4.3. Vehicle Dynamics and Control. Operating a forklift is fundamentally different from driving a standard passenger vehicle due to steering configurations, weight distribution, and braking dynamics.

4.3.1. Steering Systems. Most forklifts utilize rear-wheel steering or articulated steering.

4.3.1.1. Rear-Wheel Steering: Allows for maximum maneuverability and tight turns in warehouse environments. However, it creates a severe "tail swing." When turning, the rear of the forklift swings wide, increasing the risk of striking pedestrians or structures.

4.3.1.2. Articulated Steering: Primarily used on all-wheel-drive, all-terrain forklifts. The vehicle pivots in the middle, meaning all moving parts track in the same direction.

Figure 4.5. Forklift Steering Systems.



4.3.2. Speed and Acceleration. Speed limit's reaction time and heavily affects the operator's ability to turn safely. Rapid acceleration or "jerking" can cause mechanical failure, constitutes vehicle misuse, and can easily cause loads to fall off the forks. Operators must gradually accelerate to ensure load stability.

4.3.3. Maneuvering. Operators must keep both hands on the steering wheel during transit. During loading and unloading, the left hand is used to steer while the right hand operates the hydraulic controls. Never turn on an incline. Forklifts are highly unstable when the forks are elevated (even when empty) and tip-overs can occur easily during sharp or elevated turns.

4.3.4. Stopping and Braking. Stop the forklift as gradually as possible. Hard braking or wheel sliding can launch the load off the forks. The distance required to stop a forklift depends on three factors: Driver Perception Time, Driver Reaction Time, and Vehicle Stopping Time (which includes "brake lag" on airbrake equipped models).

4.3.4.1. Several environmental and mechanical factors dictate how quickly a forklift can stop.

Table 4.1. Braking Factors.

Factor	Impact on Braking
Surface Condition	Concrete and asphalt offer better stopping traction than unpaved surfaces.
Contaminants	Ice, snow, leaves, water, or mud severely increase stopping distance.
Road Configuration	Inclines, declines, curves, and high crowns shift vehicle weight and affect braking power.
Tire Condition	Tread depth, tire type, and proper inflation are critical for grip.
Brake Health	Type of brakes (air vs. hydraulic) and state of adjustment/repair.
Load Weight	An unloaded forklift will stop sooner and smoother than a fully loaded one.

4.4. Vehicle Inspection. A thorough vehicle inspection is the operator's first line of defense against accidents and mechanical failures. All inspections must be documented using DAF Form 1800, along with the walk-around guide located in **Attachment 2**. **Note:** If any discrepancies are found, the operator must immediately report them to the Vehicle Control Official (VCO), the supervisor, and/or Vehicle

Maintenance. Only Vehicle Maintenance can authorize the continued use of a vehicle with an active maintenance discrepancy.

4.4.1. The Seven-Step Inspection Method. To ensure consistency and guarantee no critical components are skipped, operators should utilize the Seven-Step Inspection Method detailed in **Attachment 4**. Pre-trip inspections must be conducted at the beginning of each shift.

4.4.2. Pre-Operation (Pre-Trip) Inspection. The pre-trip inspection is designed to identify leaks, damage, and missing items that could cause a breakdown or accident before the forklift is put under load.

Table 4.2. Pre-Trip Inspection Items and Description.

System / Area	Items to Inspect
Fluids & Leaks	Check for fuel, oil, coolant, hydraulic, and air leaks. Verify fluid levels are within limits: Engine oil, brake fluid, transmission fluid, antifreeze, and hydraulic fluid.
Tires & Wheels	Check proper inflation, tread depth, and look for cuts/abrasions. Inspect all wheel rims for cracks or splits, and check for loose/missing lug nuts. WARNING: Notify the VCO and Vehicle Maintenance immediately if a split rim tire is completely flat.
Mechanical & Structural	Inspect drive belt tension and fraying. Check air restriction gauges, air tanks, exhaust system, the carriage, mast tines/roller tines, and towing connections. Inspect all hydraulic hoses and cylinders for damage or leaks.
Safety & Electrical	Check battery security, fluid, damage, and corrosion. Verify operation of all wiring, lights, reflectors, mirrors, safety devices, and the horn. Ensure all required markings are visible.
Cab / Operator Area	Inspect windshield wipers/washers, heater/defroster, brake/accelerator pedal covers, cold weather aids, emergency stand-by systems, and the fire extinguisher. Refill alcohol injector reservoir (in season).

4.4.3. During-Operation Inspection. Inspection does not end once the vehicle is moving. Operators must continuously monitor the forklift's performance and immediately safely park the vehicle if a malfunction occurs.

Table 4.3. During Inspection Items and Description.

Focus Area	Operational Checks
Dashboard & Senses	Continuously monitor all gauges, warning lights, and indicators for proper operation. Listen for unusual mechanical noises, exhaust leaks, or air leaks.
Driving Controls	Test the steering wheel (and tilt control), direction control lever, gear selector, parking brake, accelerator pedal, service brakes, and the inching pedal. Test the ignition switch, battery disconnect, and fuel selector (for bi-fuel engines).

Hydraulic Controls	Verify smooth operation of the lift/lower control lever, tilt control lever, side shift (if equipped), fork positioning lever, and any specialized controls (rotate, swing, extend).

4.4.4. After-Operation Inspection and Servicing. Upon completing the mission or shift, operators must ensure the forklift is cleaned (free of dirt, excess oil, and grease) and properly refueled or charged for the next shift.

4.4.4.1. Refueling Procedures: Regardless of fuel type, operators must use designated refueling areas, turn off the engine, apply the parking brake, and strictly prohibit smoking or the use of electronic devices.

4.4.4.1.1. Diesel/Gasoline: Ensure the nozzle is seated and do not overfill.

4.4.4.1.2. Compressed Natural Gas: Ensure the refueling hose is properly and securely connected before initiating flow.

4.4.4.1.3. Propane (LPG): Turn off the fuel valve at the cylinder. Allow the engine to run until it dies to safely purge all liquid phase propane (LPP) from the lines. Once purged, disconnect and remove the cylinder.

4.4.4.2. Battery Charging Procedures: When charging electric forklifts, operators are exposed to hazardous electrical currents and corrosive battery acid. The following safety measures are mandatory:

4.4.4.2.1. Always wear appropriate eye protection when making connections.

4.4.4.2.2. Use only the manufacturer's approved charger.

4.4.4.2.3. Ensure the charging station has an accessible eye wash facility and an acid neutralizing agent available.

4.4.4.2.4. When servicing battery cells, do not overfill the batteries with water.

4.5. Vehicle Safety and Equipment.

4.5.1. Hazards and Human Factors. Operating a forklift safely requires constant attention to vehicle stability, surrounding personnel, and load security. Failing to respect these physical limits is the primary cause of industrial mishaps.

4.5.1.1. Common Mishap Types. The vast majority of forklift-related mishaps result in:

4.5.1.1.1. Dropped or Damaged Property: Caused by unbalanced loads or sudden maneuvers.

4.5.1.1.2. Personnel Injuries: Bruises, head injuries, cuts, and lacerations.

4.5.1.1.3. Fatalities: Usually resulting from vehicle roll-overs where the operator is crushed.

4.5.1.2. Mishap Causes vs. Safe Operating Behaviors. To prevent mishaps, operators must actively replace dangerous driving habits with safe, standardized procedures.

Table 4.4. Mishap Causes vs. Safe Operating Behaviors.

Avoid This (Mishap Cause)	Do This Instead (Safe Operating Behavior)
Jerky Starts, Stops, and Sharp Turns	Accelerate and brake gradually. Slow down before initiating any turn to prevent tipping.
Silent Approaches	Sound the horn at all intersections, blind corners, and when entering or exiting buildings.
High Fork Travel (Loaded or Empty)	Keep tines no higher than 6 inches above the working surface while traveling.
Unstable Loads	Ensure the pallet load is properly balanced, stacked, and that the forks are fully inserted under the load before lifting.
Striking Obstacles	Prevent the forks from striking the floor or pallets. Maintain high rear visibility to manage tail swing and avoid cutting corners too sharply.
Blind Driving	Always follow the specific direction of travel procedures outlined in paragraph 4.8.5. or utilize a designated spotter when operating in congested areas (as defined in paragraph 4.8.3.
Improper Incline Driving	Always keep the load upgrade. Drive forward up inclines, and back down declines. Never descend an incline with the load in front.
Over-speeding on Bridge Plates	Drive slowly across bridge plates and verify the plate's load carrying capacity matches or exceeds your gross vehicle weight.
Mechanical Misuse	Always fully release the parking brake before traveling. Avoid pinch points on articulated steering models.

4.5.2. Personal Protective Equipment (PPE). Operators must wear approved safety clothing and equipment to prevent injury during operations and cargo handling.

Table 4.5. Personal Protective Equipment.

PPE Item	Requirement	Reference / Standard
Safety Footwear	Must be sturdy with an impact-resistant toe.	Certified to meet or exceed ASTM F2412/F2413 standards (IAW 29 CFR § 1910.136).
Work Gloves	Must be worn during cargo loading and unloading operations.	Safety Note: Remove all rings and jewelry before putting on gloves to prevent snagging.
High-Visibility Gear	Reflective belts or vests must be worn.	Mandatory during low-visibility operations and at all times on the flightline.
Hearing Protection	Approved earplugs or earmuffs.	Required when operating gas or diesel forklifts without an enclosed cab, or when cab

		doors/windows are open.
Weather Gear	Rain gear, cold weather gear, etc.	Safety Note: Avoid operating an open-cab forklift during active rainfall.

4.6. Driving Safety and Operational Precautions. Operating a forklift safely requires strict adherence to OSHA regulations and active Risk Management (RM) processes. Operators must constantly evaluate their surroundings, maintain positive vehicle control, and remain vigilant regarding proximity to nearby personnel.

4.6.1. Pedestrian and Personnel Safety. The operator is ultimately responsible for the safety of personnel working around the forklift. Under no circumstances are passengers permitted to ride on powered forklifts. Operators must never drive a forklift directly toward anyone standing in front of a workbench or other fixed object where they could be pinned.

4.6.1.1. Personnel are strictly prohibited from standing, working, or passing under any elevated portion of a forklift, whether it is fully loaded or completely empty. To prevent severe injury or crushing, operators must keep their arms, legs, and all other appendages within the running lines of the cab or overhead guard and never place them between the uprights of the mast.

4.6.2. Environmental and Travel Rules. Operators must constantly adjust their driving habits based on the surface, obstacles, and visibility. At all cross aisles, intersections, and blind spots, operators are required to slow down, sound the horn, and refrain from passing other forklifts traveling in the same direction. If a load restricts forward vision, the operator must adhere to the mandatory direction of travel rules detailed in paragraph 4.8.5. or utilize a designated spotter when operating in congested areas (as defined in paragraph 4.8.3).

4.6.2.1. When navigating ramps or grades in excess of ten percent, loaded forklifts must always be driven with the load upgrade—meaning operators drive forward up the ramp and back down the ramp. While traversing any incline, the mast must be tilted back and the forks raised only as far as necessary to clear the road surface.

4.6.2.2. Specialized structures require extra caution. Before driving over dock boards or bridge plates, operators must ensure they are properly secured, clear of snow or ice, and rated to handle the combined weight of the forklift and cargo. Railroad tracks should be crossed diagonally wherever possible, and forklifts must never be parked closer than eight feet from the center of the tracks.

4.6.2.3. When approaching elevators, operators must drive slowly, enter squarely once the elevator car is properly leveled, and immediately neutralize the controls, shut off the power, and set the parking brakes. If operating a motorized hand truck, the unit must always enter elevators or confined areas with the load end forward.

4.6.3. Unattended Vehicle Protocol. A forklift is legally considered unattended when the operator is twenty-five feet or more away from the vehicle while it remains in view, or whenever the operator leaves the vehicle and it is completely out of sight.

4.6.3.1. When leaving a forklift unattended, the operator must fully lower the forks, neutralize the controls, shut off the power, and set the parking brakes. If the vehicle must be parked on an incline,

the drive wheels must also be chocked. If the operator dismounts but remains within twenty-five feet of the vehicle and keeps it in view, they must still lower the forks, neutralize the controls, and set the parking brakes to prevent any accidental movement.

4.7. Forklift Load Composition and Stability.

4.7.1. The Stability Triangle and Center of Gravity. Unlike standard passenger vehicles, forklift stability relies on a three-point suspension system known as the stability triangle. This imaginary triangle is formed by the two front drive wheels and the center pivot point of the rear steering axle. To prevent tipping, the combined center of gravity—which is the shifting balance point of the forklift's weight plus the cargo's weight—must remain entirely within this triangle.

4.7.1.1. If the load is too heavy, the center of gravity shifts forward out of the triangle, causing a forward tip-over. If an operator turns too sharply or drives too fast, the center of gravity shifts laterally out of the triangle, resulting in a side roll-over. Because forklifts are longer than they are wide, the stability triangle is narrowest on the sides, making lateral side-tips far more common than forward-tips during improper handling.

4.7.2. Key Stability Terminology. To safely operate a forklift, drivers must understand the physical terms that govern stability.

4.7.2.1. The center of gravity is the exact point on an object at which all of its weight is concentrated. To balance this, the counterweight is a heavy mass built into the rear of the forklift's structure to offset the load and maximize resistance to tipping.

4.7.2.2. The fulcrum point is the axis of rotation—typically the front axle—where the vehicle will pivot if it tips forward.

4.7.2.3. The load center is the horizontal distance from the vertical face of the forks to the load's center of gravity.

4.7.2.4. Understanding these factors determines the vehicle's longitudinal stability, which is its resistance to tipping forward or backward, and its lateral stability, which is its resistance to tipping sideways.

Figure 4.6. Center of Gravity.

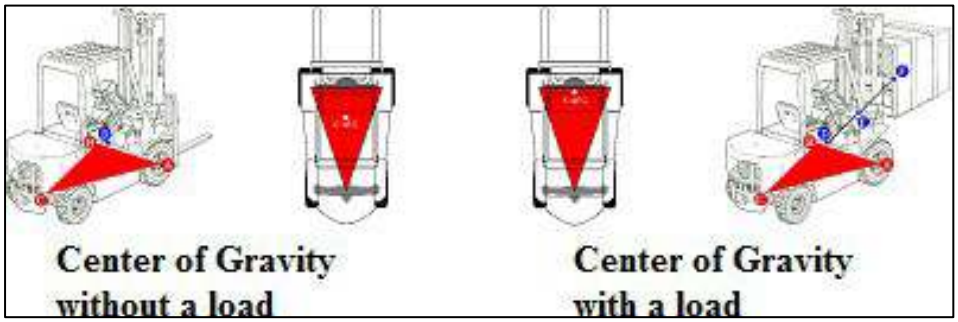
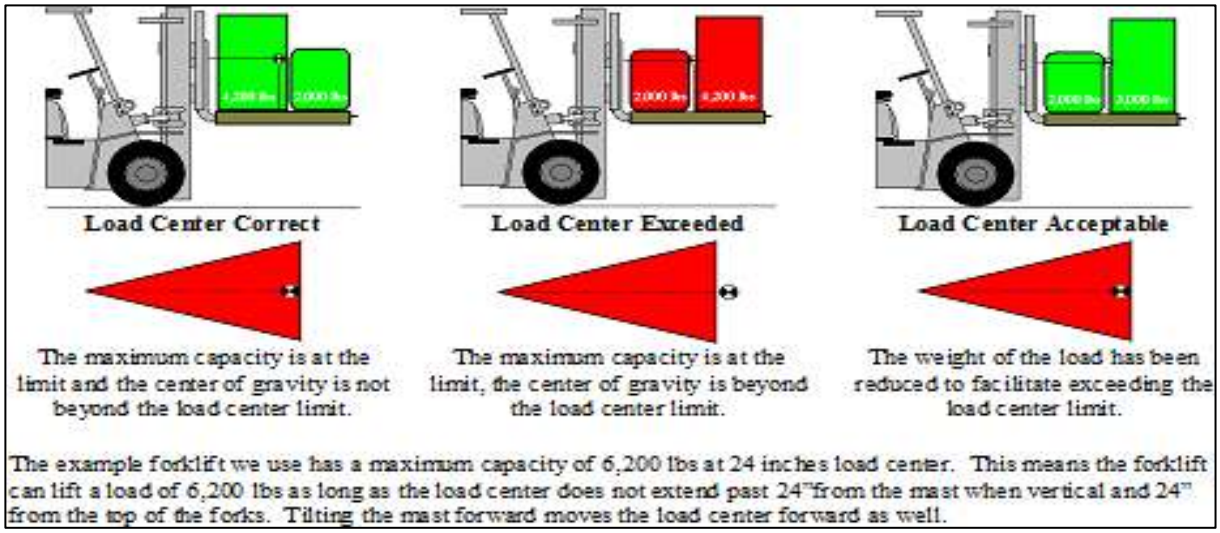


Figure 4.7. Forklift Load Center.



4.8. Forklift Loading and Unloading. Once an operator engages a load, they assume full responsibility for its safe transport. Operators must check the vehicle data plate to verify that the cargo weight is within the rated capacity at that specific load center.

4.8.1. When engaging a load, the forks must be placed under the cargo as far as possible until the load is flush against the mast. The mast should then be carefully tilted backward just enough to stabilize the load against the backrest.

4.8.2. Operators must exercise extreme caution when tilting, especially when high-tiering. Tilting an elevated load forward is strictly prohibited unless the cargo is in a deposit position directly over a rack or stack.

4.8.2. Preparing and Securing Loads. Cargo comes in all shapes and sizes, and operators must ensure it is ready for transport. Whenever possible, cargo should be consolidated onto standard pallets, such as skids, 463L pallets, or self-contained kits, to establish a stable base.

4.8.2.1. All loads must be tightly packed to prevent shifting and secured to the pallet using cargo nets, straps, chains, or heavy plastic wrap before lifting. If moisture-sensitive cargo must be moved outdoors during inclement weather, the load should be covered with a tarpaulin.

Figure 4.8. Forklift Load Styles.

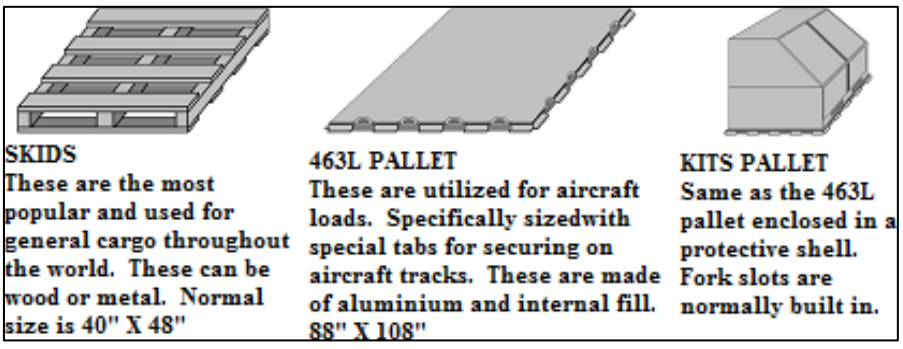
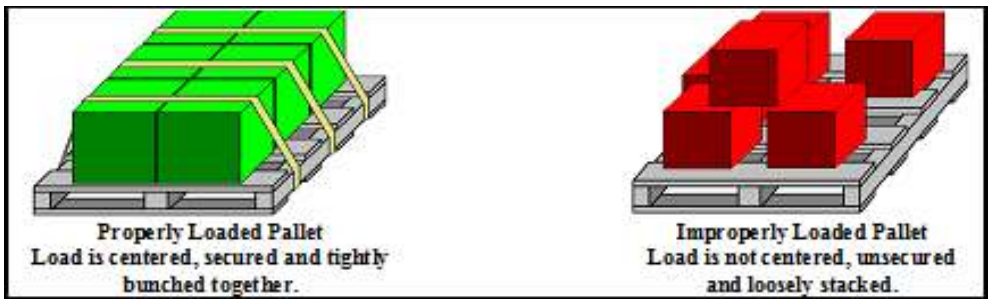


Figure 4.9. Properly/Improperly Loaded Pallet.



4.8.3. Spotter Utilization and Communication. Utilizing a spotter is necessary during loading, unloading, and maneuvering in tight or congested areas. However, a spotter is not required for standard forklift, straight-line transit in reverse across open areas where the operator maintains a clear line of sight behind them and when not nearby aircraft. Spotters are also required for any forward travel under 50 feet where the operator's view is obstructed. Congested areas are defined as areas where objects are within 15 feet of forklift operations and obstructed view is when operator's forward view is obstructed by the carriage, forks, or load, preventing them from seeing a 5-foot-tall person or hazard located 10 feet directly in front of the vehicle. However, a spotter must not be used as a substitute for mandatory reverse travel during long-distance transits (over 50 feet) with an obstructed view.

4.8.3.1. The most critical rule of spotter operations is that if visual contact with the spotter is lost, or if the operator is ever in doubt of a hand signal, they must stop the forklift immediately. Hand signals must be clear and standardized to ensure safe, flawless communication between the spotter and the driver. See **Figures 4.10. – 4.17.**

Figure 4.10. Forklift Hand Signals – Close/Open Fork Tines.



Figure 4.11. Forklift Hand Signals – Side Shift Right/Left.



Figure 4.12. Forklift Hand Signals – Raise/Lower Carriage.



Figure 4.13. Forklift Hand Signals – Tilt Mast Forward/Back.



Figure 4.14. Standard Spotting Hand and Arm Signals – Come Towards the Spotter.



Figure 4.15. Standard Spotting Hand and Arm Signals – Move Away from the Spotter.

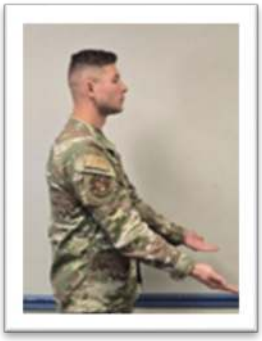


Figure 4.16. Standard Spotting Hand and Arm Signals – Right/Left Turn.



Figure 4.17. Standard Spotting Hand and Arm Signals – Slow Down, Stop, and Cut-off Engine.



4.8.4. Picking Up Loads. When approaching a load to pick it up, the operator must move in slowly, ensuring the forklift is square and centered with the target cargo. The forks must be spread as wide as possible to match the width of the load, which maximizes lateral stability on the tines. Ensure the forks are at the proper height for engagement and then drive forward slowly until the tines are entirely under the load.

4.8.4.1. During this approach, operators must be careful not to over-extend the forks, which could accidentally pierce and damage cargo positioned directly behind the target load. Once the forks are fully inserted, the operator should lift the load slightly, inch back, lower the cargo, and fully rest the load against the carriage. Before moving, raise the load just enough to clear the driving surface and tilt the mast back slightly to secure the cargo against the load backrest. If picking up from a stack, slowly back the forklift completely clear of the racking structure before lowering the carriage to a safe travel height.

4.8.5. Transporting Loads. When transporting cargo, the mast must remain tilted back just enough to stabilize the load, and the forks should never be raised higher than necessary to achieve adequate ground clearance. The operator's forward visibility is the absolute governing factor for direction of travel.

4.8.5.1. Direction of Travel. The forklift must be driven in reverse if any of the following conditions are met:

4.8.5.1.1. Severe Obstruction: The operator's forward view is obstructed by the carriage, forks, or load, preventing them from seeing a 5-foot-tall person or hazard located 10 feet directly in front of the vehicle.

4.8.5.1.2. Bulky Load: The cargo extends wider than the forklift carriage or higher than the operator's dashboard when transported at the standard 6-inch height.

4.8.5.1.3. Long Distance Transit: The travel distance exceeds 50 feet or involves moving between distinct work areas (e.g., from a warehouse to the flightline). For transits under 50 feet with an obstructed view, a designated spotter is required.

4.8.5.2. All-Terrain (AT) Forklifts. When operating all-terrain forklifts the standard procedure is to travel forward due to elevated cab design. However, if an oversized load causes severe obstruction as defined above, reverse travel is required.

4.8.5.3. Operators must avoid any sudden starts, stops, or sharp turns. For over-the-road forklift transit, the palletized load must be physically strapped and secured to the forklift frame.

4.8.6. Unloading and Stacking Procedures. To unload cargo onto the floor, the operator must level the mast with the ground and slowly lower the carriage. It may be necessary to apply a slight forward tilt to allow the forks to slide out smoothly as the vehicle backs away. When backing out, operators must ensure the rear heels of the forks do not drag along the floor, as this can severely damage or break the carriage keepers over time.

4.8.6.1. To place a load onto an existing stack, the operator should slowly raise the carriage while inching the vehicle into position, exercising extreme caution to avoid colliding with adjacent cargo. Once the load is squared directly over the stack, level the mast and lower the forks just enough to transfer the weight of the load onto the stack and free the tines. Operators must be careful not to drag the forks against the pallet as they back away, and they must immediately lower the carriage to a safe travel height as soon as the forklift clears the stack.

4.8.6.2. When unloading specialized cargo platforms such as 463L or KITS pallets, the load must always be placed on wood shoring (dunnage). Shoring protects the pallets from ground moisture and dirt, while providing the necessary clearance for the forks to slide in and out. Operators must utilize at least three pieces of evenly spaced dunnage for each pallet to prevent the platform from warping under its own weight.

4.9. Forklift Operation

4.9.1. Operating Inside and Outside of Warehouses. Operating a forklift within and around warehouse facilities requires strict situational awareness and adherence to speed and clearance protocols. Operators must maintain a maximum speed of 5 miles per hour (mph) indoors. When approaching warehouse openings or adjacent rooms, operators must come to a complete stop between 5 and 10 feet away and sound the horn to alert unseen personnel.

4.9.1.1. When navigating warehouse ramps or loading docks, always downshift the forklift to "L" (Low) prior to descending or ascending, ensuring the correct gear is selected for safe operation in and out of buildings. Additionally, environmental hazards must be managed: fuel-powered forklifts (gas, diesel, propane, CNG) may only operate in approved, well-ventilated warehouses, and only specifically approved forklifts are authorized in warehouses storing hazardous materials.

4.9.2. Forklift Attachments Use. Two of the most common attachments encountered on a forklift

are rollerized tines and personnel lift platforms.

4.9.2.1. Rollerized Tines. Rollerized tines are specialized attachments designed to expedite the unloading of rollerized trailers, docks, and aircraft during contingency operations. While these accessories significantly simplify cargo handling, they introduce distinct hazards that require strict operational discipline. When issued, operators must integrate the tines into their pre-use inspections and schedule them for preventive maintenance with Transportation.

4.9.2.1.1. They must only be mounted on forks matching their exact size and weight capacity and must never be used as makeshift fork extensions or to lift SKID-type pallets. When stored, keep them flat in designated work areas so they do not present a tripping hazard.

Table 4.6. Mounting and Operational Steps.

Mounting & Operational Steps	Action
Mounting	Lay the tines flat on a level surface and slowly drive into them. Do not manually lift the tines onto the forks.
Securing	Ensure tines are mechanically locked to the carriage. Any cargo loaded onto them must be strapped securely to the forklift itself.
Transport	Use rollerized tines only when absolutely necessary and limit transit to the shortest distance possible.

4.9.2.2. Personnel Lift Platforms. Whenever a forklift is equipped with vertical or horizontal platform controls, strict precautions must be enforced to protect elevated personnel. A certified safety platform must be firmly secured to the lifting carriage and/or forks. Operators must ensure there is a reliable means for personnel on the platform to shut off power to the forklift in an emergency, and overhead protection from falling objects must be provided based on the operating environment.

4.9.3. Parking Procedures. Parking a forklift safely requires adherence to specific shut-down sequences to prevent property damage or runaway vehicles. Forklifts must never be parked where they limit access to fire aisles, stairways, or fire equipment.

4.9.3.1. The procedures differ slightly depending on whether the operator is in a standard yard/warehouse or operating under strict flightline regulations.

Table 4.7. Standard v. Flightline Parking Procedures.

Step	Standard Parking Procedure	Flightline Parking Procedure (DAFI 13-213)
1	Find a level area.	Position on designated flightline parking spot.
2	Place transmission in Neutral and apply the Parking Brake.	Place transmission in Neutral and apply the Parking Brake.
3	Lower carriage completely; tilt mast	Lower carriage completely; tilt mast

	forward until fork tips touch the ground.	forward until fork tips touch the ground.
4	Idle engine: 5 mins for turbo engines; per manufacturer for non-turbo.	Turn ignition switch to the OFF position.
5	Turn off all lights and accessories.	Chock the drive wheels (pre-positioned chocks).
6	Turn ignition switch to OFF (and master switch, if applicable).	Leave key in the ignition.
7	Chock the drive wheels if parked on an incline.	Ensure beacon light and/or flashers are functional.

4.9.4. Flightline Operations. All airfield forklift operations must comply with DAFI 13-213, *Airfield Driving*. Flightline training and licensing are conducted separately from basic forklift qualification.

4.9.4.1. When operating on the flightline, drivers must not exceed 5 mph when maneuvering around aircraft. Unless concurrent servicing is explicitly authorized, never load or unload cargo while the aircraft is being serviced with fuel or oxygen. Always utilize a designated spotter when approaching or backing away from an aircraft and ensure pre-positioned wheel chocks are placed between the aircraft and the forklift during transfer operations.

4.9.5. Fire Extinguishers & Hazardous Materials. Fire Extinguisher Readiness: Most forklifts are equipped with mounted fire extinguishers, requiring operators to undergo annual training scheduled by the supervisor or Squadron VCO through the local Fire Department. Prior to operation, operators must inspect the extinguisher to ensure it has a current inspection tag, the pressure needle is in the green, and the unit is securely mounted. (Note: Refer to *DAFMAN 91-201* for specific extinguisher quantity and rating requirements when transporting explosives).

4.9.5.1. Hazardous Materials (HAZMAT): Personnel tasked with transporting explosives or hazardous materials require separate, specialized training. Trainees must successfully complete this instruction prior to handling any hazardous cargo.

4.9.6. Workplace Policies and Procedures. Trainees must fully understand all local wing, squadron, and shop operating procedures regarding forklift operations. If operating at a temporary duty (TDY) location, operators have a responsibility to proactively review and adhere to the specific local policies and host-installation airfield driving instructions before operating any equipment.

Chapter 5

EXPLANATION AND DEMONSTRATION

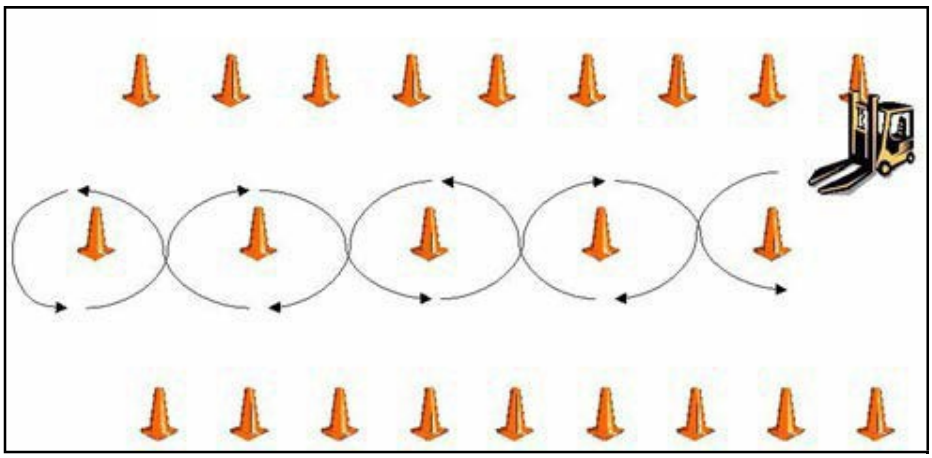
5.1. Instructor Preparation and Safety Protocols. Prior to instruction, the instructor must establish a safe training location, reserve a forklift, obtain the manufacturer's operator's manual, and ensure the trainee has completed a DAF Form 171. Both the instructor and trainee must adhere strictly to safety protocols, including the removal of all jewelry and identification tags, the use of wheel chocks when the forklift is parked (if required), and the mandatory use of seat belts. The instructor should conduct an initial walk-around to familiarize themselves and the trainee with all warning labels and signs. Throughout the demonstration, the instructor must emphasize practicing forklift safety: observing speed precautions, keeping loads within rated capacity, maintaining proper overhead and side clearances, avoiding sudden stops, and never extending hands or arms through the mast. The instructor will also demonstrate traveling with forks raised no more than 6 inches from the floor and driving in reverse when carrying bulky loads or when the mast obstructs the field of vision.

5.1.1. Personal Protective Equipment (PPE) and RM. Instructors and trainees must wear appropriate PPE, including sturdy safety footwear with impact-resistant toes. Heavy-duty gloves must be worn during cargo loading and unloading. During the demonstration, the instructor will actively model the basic Air Force Risk Management (RM) process by identifying hazards, assessing hazards, developing and implementing controls, and supervising and evaluating the operation.

5.2. Operator Maintenance and Vehicle Pre-Trip Inspection. The instructor will accomplish a comprehensive vehicle inspection alongside the trainee, documenting it on the DAF Form 1800. The instructor will demonstrate the mandated seven-step inspection method **(Attachment 4)** and utilize the inspection guide **(Attachment 2)** to ensure all areas of the forklift are covered in addition to the operational guidelines.

5.3. Operational Demonstration. Utilizing the vehicle data plate and the manufacturer's operator's manual, the instructor will explain the forklift's capabilities and stability principles, emphasizing the center of gravity and the stability triangle. Throughout the demonstration, the instructor will allow for questions and repeat actions as needed. The instructor will demonstrate the proper use of the parking brake (air or manual), brake pedal, inching pedal (if equipped), transmission gearshift levers, speed selection, and directional controls.

Figure 5.1. Forklift Maneuverability Course Example.



5.3.1. Steering and Hydraulic Controls. Steering systems vary across models, so the instructor must demonstrate maneuverability specific to the vehicle (e.g., front wheel, rear wheel for standard 4-10K forklifts, or articulated

steering for 10-13K AT forklifts). The instructor will navigate a suggested maneuverability course while keeping the forks low to the ground. Additionally, the instructor will show the operation of all hydraulic control levers, including lift/lower controls (standard and articulating reach), tilt control, side shift, fork positioning levers, pivot locks, and any other equipped controls (rotate, swing, extend).

5.3.2. Loading, Unloading, and Spotter Procedures. The instructor will go over standard hand signals and have the trainee serve as the spotter, ensuring the spotter is kept in sight at all times. For loading, the instructor will demonstrate approaching slowly, squaring and centering the forklift with the load, spreading the forks to the correct width, inserting forks entirely under the load, and lifting and tilting the mast slightly back before slowly backing away. For unloading on the floor or onto stacked cargo, the instructor will demonstrate positioning the load level, lowering it, tilting the mast forward slightly to clear the load, and backing out slowly. The instructor will also demonstrate the use of shoring (dunnage) for 463L pallets.

5.4. Post-Operation Demonstration. Following the driving and load handling portion, the instructor will demonstrate after-operation procedures. This includes cleaning the vehicle, refueling, and executing the manufacturer's shut-down procedures. The instructor will park the forklift in a level area, place the transmission in neutral, apply the parking brake, and completely lower the carriage until the fork tips touch the ground. The instructor will demonstrate proper engine cool-down (e.g., letting a turbo-charged engine idle for 5 minutes), turning off all lights and the master switch, and chocking the drive wheels if parked on an incline or flightline. Finally, the instructor will drain the air tanks (if applicable), perform a post-operation walk-around, and annotate any discrepancies on the DAF Form 1800 before concluding with time for final questions.

Chapter 6

TRAINEE PERFORMANCE AND EVALUATION

6.1. Trainee Performance and Instructor Responsibilities. The instructor's primary directive during hands-on training is to ensure absolute safety at all times. The instructor must immediately stop training if any safety violations occur and only resume once the trainee fully understands how to avoid repeating the infraction. Before allowing the trainee to operate the forklift, the instructor must ensure that both parties remove all jewelry and identification tags, use wheel chocks (if required) when parked, wear seat belts, and properly adjust the driver's seat and mirrors. Additionally, both the instructor and trainee must wear sturdy safety footwear with impact-resistant toes, heavy-duty gloves during all cargo loading and unloading operations, proper hearing protection, and a highly visible reflective belt or vest during low-visibility periods. The instructor must ensure the trainee pays close attention to all warnings and cautions in the manufacturer's manual, is fully briefed on the specific driving scenarios they are to execute and participates in ongoing during- and after-action reviews to correct errors or re-demonstrate complex tasks.

6.1.1. Hands-On Trainee Practice. Under direct supervision, the trainee must execute all phases of operator maintenance, conduct a comprehensive pre-inspection and during-operation inspections while explaining each checklist item aloud to the instructor. The trainee may utilize the inspection guide in **Attachment 2** as a reference, and they must properly document all findings on DAF Form 1800. Once inspections are complete, the trainee will practice basic forklift operations forward and backward, navigating a established road course complete with turns and stop signs. When backing, the instructor or another trainee must act as a spotter, and the operator must maintain visual contact with them at all times. This fundamental driving practice must continue until the trainee demonstrates complete operational proficiency.

6.1.2. Maneuverability and Cargo Handling Exercises. Once basic driving is mastered, the trainee will progress to more complex maneuverability and cargo handling exercises. Using a recommended maneuverability course layout (See **Figure 5.1.** Forklift Maneuverability Course Example), the trainee will practice picking up and transporting loads. This process begins by stacking four to six empty wooden pallets (or 463L pallets if training on a 10K 463L forklift), loading them one at a time, navigating the course, and restacking them at the destination. Empty pallets are utilized initially to maximize visibility so the trainee can learn proper approach and fork engagement. Once proficient, the course can be rearranged or modified to include forward and reverse operations. The trainee will then transition to actual training cargo, practicing loading, transporting, and unloading onto a trailer or dock. Instructors should continually modify the course layout and introduce realistic mission-essential scenarios to build the trainee's adaptability. Each session must conclude with the trainee performing a thorough after-operation inspection and executing the correct parking and shutdown procedures.

6.2. Performance Evaluation and Certification. Once the instructor determines the trainee is ready, they will transition to the formal performance evaluation. The goal of this evaluation is to certify that the trainee understands all safety precautions, can perform proper operator maintenance IAW applicable technical orders, and can safely and proficiently operate the forklift to meet mission requirements and OSHA/AFOSH standards.

6.2.1. The evaluation will be formally documented using the Performance Test Checklist **(Attachment**

3). Prior to beginning, the instructor and trainee will thoroughly review the checklist together, and the instructor will address any remaining questions. If available, the training forklift should be marked with a magnetic "Driver-in-Training" sign.

6.2.2. Evaluation Standards and Retraining Procedures. During the evaluation, the trainee is not allowed any instructor assistance. The instructor will observe and score the trainee's performance against the tasks listed on the checklist. If the trainee fails to meet the standard on any task, it will be marked as a "No-Go." Following the evaluation, the instructor will conduct an after-action review, re-demonstrate the failed tasks, and provide targeted coaching. The trainee must practice these areas until proficient, at which point a re-evaluation will be conducted on the "No-Go" items.

6.2.3. Certification and Recertification Requirements. A trainee must successfully pass all items on the Performance Test Checklist to receive their initial certification. To maintain qualification on Powered Industrial Trucks, this performance test must be successfully re-accomplished every three years (*Ref: paragraph 2.3.1*.).

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFI 24-301, *Ground Transportation,* 9 January 2026

DAFI 13-213, *Airfield Driving*, 20 November 2023

DAFMAN 91-201, *Explosive Safety Standards*, 18 June 2025

OSHA Directive, Enforcement and Compliance (CPL) 02-01-028 (CPL 2-1.28A), *Compliance Assistance for the Powered Industrial Truck Operator Training Standards*

Regulations (Standards - 29 Code of Federal Regulations (CFR)), Part 1910 - Occupational Safety and Health Standards, Subpart N – *Material Handling and Storage*

Title 29 CFR Part 1910, *Occupational Safety and Health Standards, Subpart N – Material Handling and Storage*

Adopted Forms

DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Drivers*, 1 November 2018

DAF Form 847, *Recommendation for Change of Publication*

DAF Form 1800, *Operator's Inspection Guide and Trouble Report*

Abbreviations and Acronyms

ABS—Anit-lock Braking System
AF—Air Force
AFIMSC—Air Force Installation Mission Support Center
AFMAN—Air Force Manual
AFOSH—Air Force Occupational Safety and Health
AT—All Terrain
CFR—Code of Federal Regulations
DAF—Department of the Air Force
DAFI—Department of the Air Force Instruction
DoD—Department of Defense
HAZMAT—Hazardous Materials
IAW—In Accordance With
LPG—Liquefied Petroleum Gas
LPP—Liquid Phase Propane
MPH—Miles per Hour
OSHA—Occupational Safety& Health Administration
PPE—Personal Protective Equipment
PSI—Pounds per Square Inch

QTP—Qualification Training Package
RM—Risk Management
TDY—Temporary Duty
TO—Technical Order
VCO—Vehicle Control Official

Attachment 2

VEHICLE INSPECTION GUIDE

A2.1. Desired Learning Outcome.

A2.1.1. Understand the safety precautions to be followed before, during, and after operation of the forklift.

A2.1.2. Understand the purpose of the forklift and their role in the mission.

A2.2. Inspection During-Operations. The operator must ensure the following items are checked after starting the forklift and during operations.

A2.2.1. Gauges and Warning Lights. Most warning lights will show when the ignition is in the start position to show proper operation of the bulb. Some gauges will "peg" to one side showing operation.

A2.2.2. Once the engine has started these warning lights should go out and gauges will go to appropriate readings. It is important the operator becomes familiar with the proper operating readings of gauges; specific meaning of warning and indicator lights and what to do if indications are warning of a malfunction on the forklift the operator will be operating.

A2.2.3. Some forklifts will have all these items, and some will be limited. Along with indicators and gauges; warning lights may service the same function, however placed on the forklift as an added precaution from operator error. Some generic forklift warning lights, indicators and gauges are shown as follows:

Figure A2.1. Forklift Warning Lights.

Display:	Indicates:	Response:
	Indicates low brake pressure.	Wait for pressure to build up before moving forklift.
	Indicates low engine oil pressure	Check engine oil level, fill to required level.
	Indicates machine requires servicing soon.	Discontinue use and contact VCO for support.
	Indicates parking brake is engaged.	Will remain on until parking brake is released.
	Indicates an electrical malfunction.	Discontinue use and contact VCO for support.
	Indicates the transmission filter restriction.	Discontinue use and contact VCO for support.
	Indicates the hydraulic oil filter restriction.	Discontinue use and contact VCO for support.
	Indicates the air filter is restricted.	Remove and shake out air filter if dry type.
	Indicates intake air restricted.	Remove and shake out air filter if dry type.
	Indicates transmission overheating.	Stop use of forklift and let transmission cool down.
	Indicates alternator malfunction.	Check drive belt tension.
	Indicates coolant level is low.	Check coolant level and fill.
	Indicates seatbelt not in use.	Fasten seat belt.

Figure A2.2. Forklift System Gauges.

Display:	Gauge:	Description:	Reading Evaluation:
	Hour Meter	Operates when the key switch is ON and the engine is running.	Periodic maintenance and vehicle rotations are based on these hours.
	Ammeter	Actuates once the ignition switch is ON, shows the amount of AMPS the alternator is putting out.	Normal ammeters indicate red or green. If the gauge shows red check for alternator belt tension first.

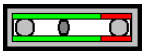
	Transmission Oil Temperature	Actuates once the ignition switch is ON, shows reading of transmission fluid temperature.	During heavy movement or driving on inclines the transmission temperature will increase. If it shows over heating discontinue use and let the vehicle cool. Check fluid level.
	Coolant System Temperature	Actuates once the ignition switch is ON, shows reading of engine coolant temperature.	During heavy movement or driving on inclines the coolant temperature will increase. If it shows over heating discontinue use and let the vehicle cool. Check coolant level.
	Engine Oil Pressure	Actuates once the ignition switch is ON and the engine is running, shows reading of engine oil pressure	When the engine is cold and oil viscosity is higher, the gauge should read a high oil pressure. 40 to 60 psi is a normal operating range. If oil pressure drops below required level, discontinue use; check engine oil level.
	System Air Pressure	Actuates once the ignition switch is ON, shows reading of system air pressure.	To release the parking brake, system air pressure normally requires 60 psi. Some systems are split and the gauge will have two indicators. If air pressure drops below required level the parking brakes will apply.
	Hydraulic Filter Restriction	Actuates once hydraulic fluid flow has started, normally once the engine is running.	Sometimes placed in the cab and other times on the hydraulic system by the hydraulic filter. If the gauge shows in the red, discontinue use and contact the VCO.
	Air Filter Restriction	This indicator holds the highest restriction reading until manually reset.	Placed in the cab or on the engine air intake system. If the gauge shows red, remove and shake out air filter if dry type. Reset the indicator by pushing the button on the top.

Figure A2.3. Forklift System Indicators.

Display:	Gauge:	Description:	Reading Evaluation:
	Glow Plug Light	Actuates when the glow plug system is in use.	Depending on the system, glow plugs actuate automatically or when manually applied. On diesel engines, for initial start utilize the glow plugs.

A2.3. Pre-Inspection. The following are items to be inspected for proper operation before operating a forklift. Numbers correlate with **Figure A2.4.**, which can be used as a forklift inspection guide for the trainee.

A2.3.1. Cleanliness/Damage/Missing Items.

A2.3.2. Leaks (fuel/oil/coolant/hydraulic/air).

A2.3.3. Fluid Levels; ensure level is within limits:

A2.3.3.1. Engine oil. **(1)**

A2.3.3.2. Brake fluid. **(3)**

A2.3.3.3. Transmission fluid. **(4)**

A2.3.3.4. Hydraulic fluid. **(8)**

A2.3.4. Battery; security, fluid damage and corrosion. **(2)**

A2.3.5. All wheel rims (bents/dents/cracks, etc.); check for loose lug nuts. **(6)**

A2.3.6. Drive belts; tension and fraying. **(7)**

A2.3.7. Air restriction gauge. **(9)**

A2.3.8. All tires; check for proper inflation and tread. **(11)**

A2.3.9. Carriage. **(12)**

A2.3.10. Hydraulic hoses/cylinder (damage).

A2.3.11. Wiring/lights/reflectors/mirrors.

A2.3.12. Safety devices.

A2.3.13. Mast tines/roller tines.

A2.3.14. Towing connection.

A2.3.15. Markings – check visibility.

A2.3.16. Horn operation.

A2.3.17. Windshield wipers/washers.

A2.3.18. Heater/defroster.

A2.3.19. Exhaust system.

A2.3.20. Brake and accelerator covers.

A2.3.21. Air tanks.

A2.3.22. Cold weather aids.

A2.3.23. Emergency stand-by system.

A2.3.24. Refill alcohol injector (in season).

A2.3.25. Fire extinguisher(s).

Figure A2.4. Forklift Inspection Guide (1/2).

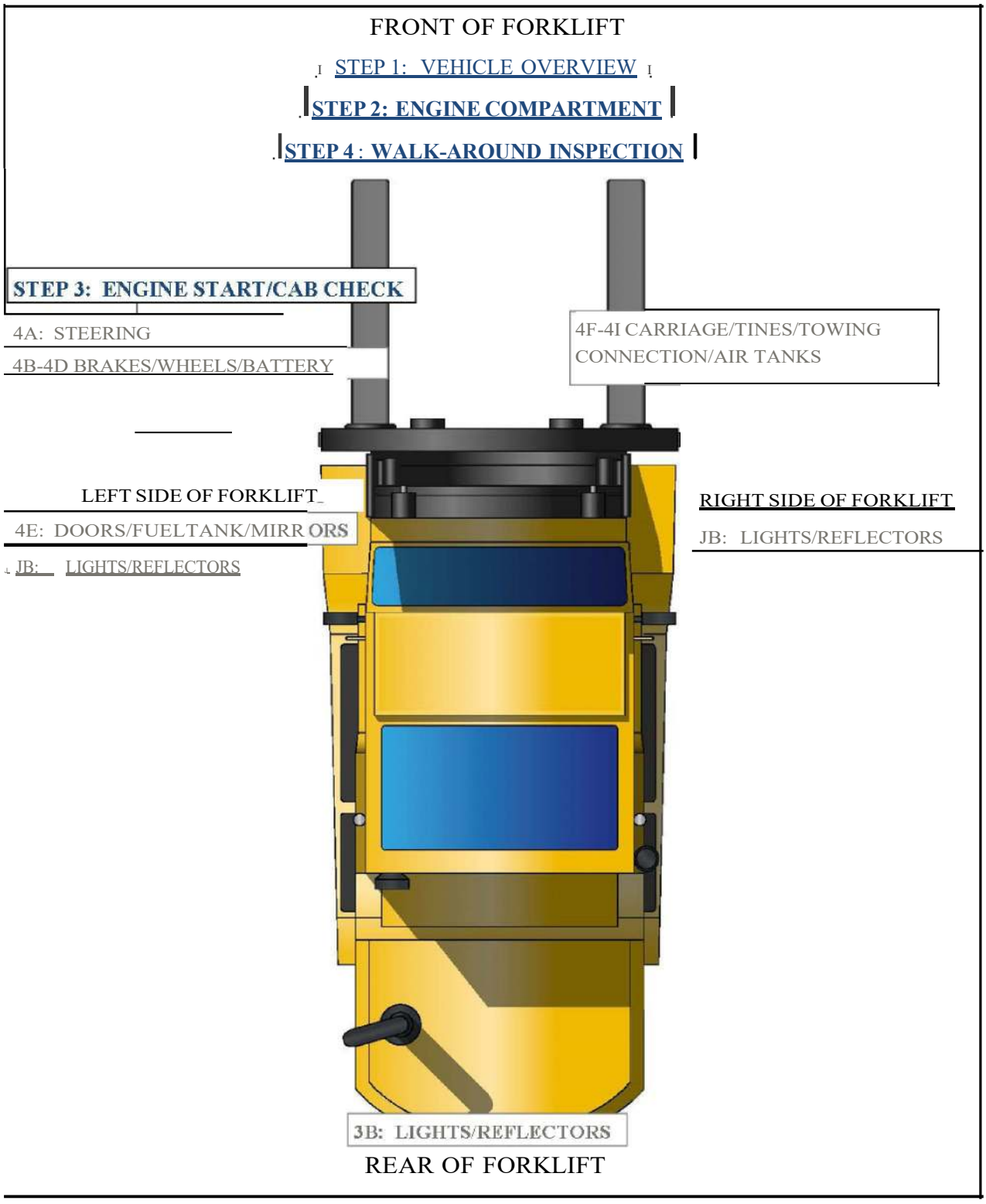
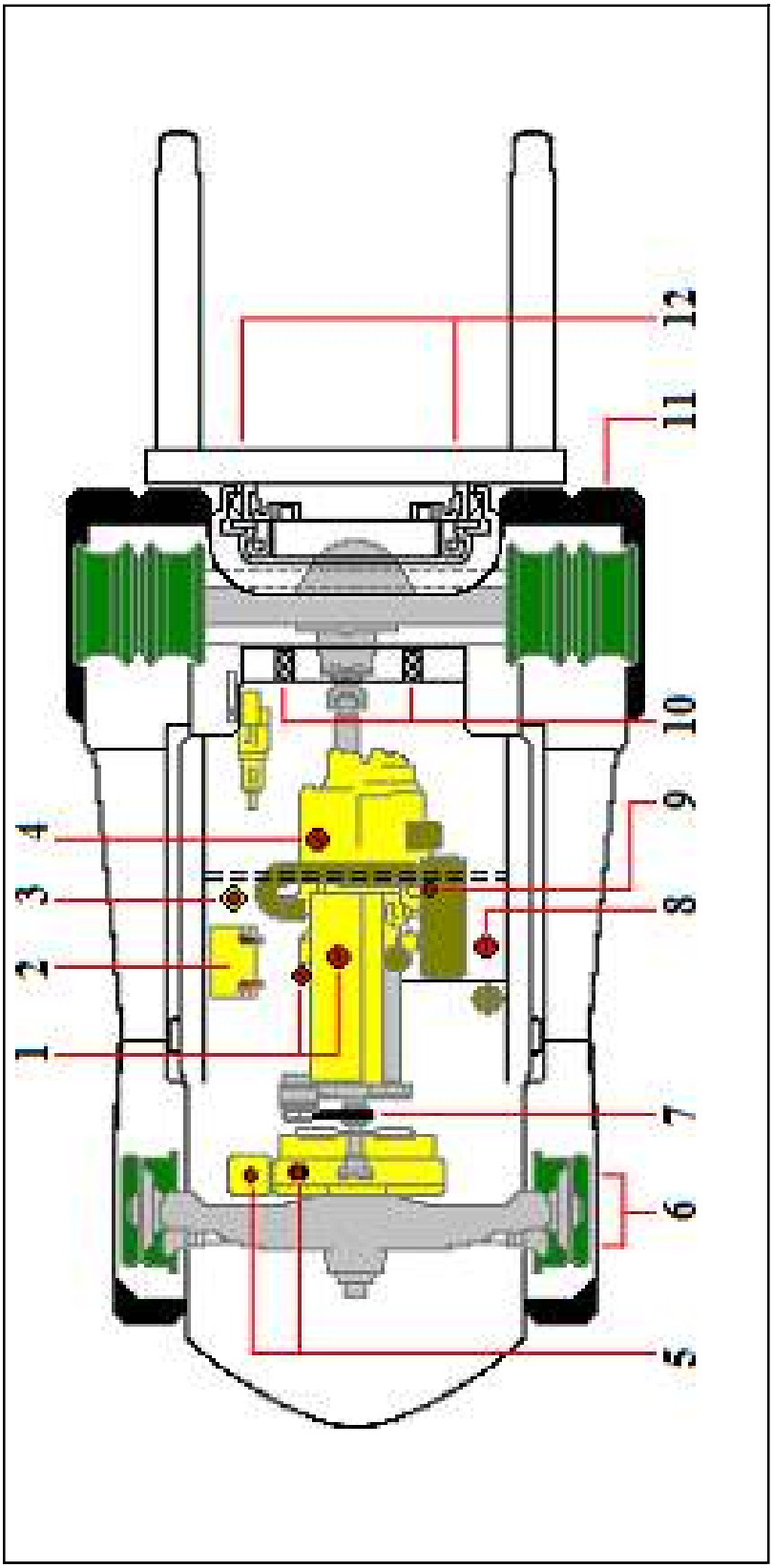
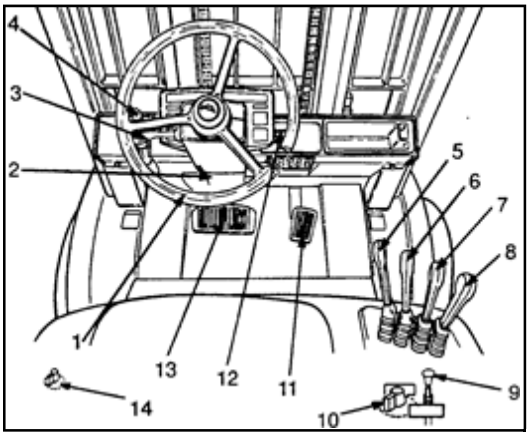


Figure A2.5. Forklift Inspection Guide (2/2).



A2.4. Forklift Controls. (See) **Figure A2.6.** illustrates the controls the operator might find on a forklift, the numbers in the figure correspond with the discussion that follows. Numbers align with the diagram.

Figure A2.6. Forklift Controls.



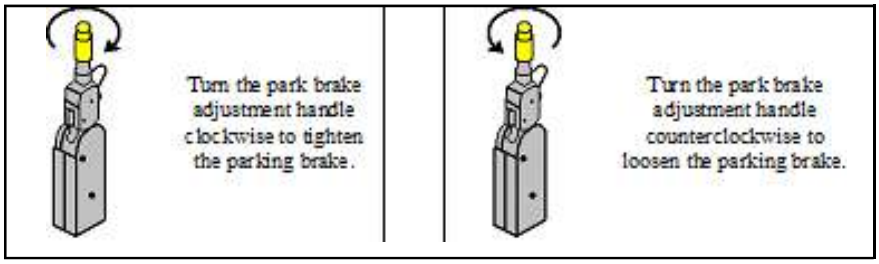
A2.4.1. Steering Wheel, controls position of steering wheels. **(1)**

A2.4.2. Steering Wheel Tilt Control, allows operator to adjust tilt of steering wheel. **(2)**

A2.4.3. Direction Control Lever. Forklifts can have from one gear in forward and reverse to four gears in each direction. It is always best to start off in the lowest gear selection and work the gears as needed. Always bring the forklift to a complete stop with the brakes before shifting transmission direction. Utilize lower gears to drive down an incline, this slows the forklift down using the engine and transmission and takes less effort when braking. **(3)**

A2.4.4. Parking Brake Control, engage the parking brake any time the forklift is parked or waiting. If the park brake does not hold when parking, adjustment is necessary. Parking brakes should hold when the forklift is sitting level, and the transmission is in reverse with engine at idle speed. Most hand lever park brakes can be adjusted by turning the end cap handle clockwise to tighten and counterclockwise to loosen. For proper adjustment, apply the service brake and place the forklift in reverse, adjust the park brake until forklift is held in place solely by the parking brake. Do not overtighten the parking brake adjustment. When adjustment is at the maximum capacity and the forklift still does not move, contact the VCO, the supervisor, and/or vehicle maintenance for repairs. **(4)**

Figure A2.7. Parking Brake Control.



A2.4.5. Hydraulic Control Levers. These controls can be either lever type or joystick actuated. By manipulating these controls, the hydraulics will move the load-carrying device. Normal functions are lift, lower, and tilt. Additional functions can be fork spread, side shift, rotate, swing, and extend. Standard forklifts normally utilize single acting hydraulic cylinders with the lift function; meaning gravity lowers the forks. Dual acting cylinders are used on all-terrain forklifts, these cylinders use hydraulic pressure to lower the forks, be careful not to dig into the driving surface by lowering too far. The following are the most common hydraulic controls on a forklift (it is recommended the operator refers to the operator's manual for proper operation of the controls). **(5-8)**

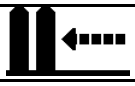
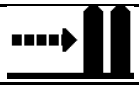
A2.4.5.1. Lift/Lower Control Lever **(5)**:

	Push the lever forward in order to lower the forks.
HOLD	The lever will return to the HOLD position when released.
	Pull the lever backward in order to raise the forks.

A2.4.5.2. Tilt Control Lever **(6)**:

	Move the lever forward in order to tilt the forks down.
HOLD	The lever will return to the HOLD position when released.
	Move the lever backward in order to tilt the forks back.

A2.4.5.3. Side Shift **(7)**:

	Move the lever forward to shift the carriage to the left.
HOLD	The lever will return to the HOLD position when released.
	Move the lever backwards to shift the carriage to the right.

A2.4.5.4. Fork Positioning Lever **(8)**:

	Move the lever to the right and forward to spread the forks.
HOLD	The lever will return to the HOLD position when released.
	Move the lever to the right and backward to close the forks.

A2.4.5.5. Off/On/Enrich Knob. **(9)**

A2.4.5.6. Battery Disconnect Switch, for routine and emergency use. Disconnects the battery and removes power from all electrical circuits for non-use times and emergency electrical situations. **(10)**

A2.4.5.7. Accelerator Control Pedal, controls engine rpm. **(11)**

A2.4.5.8. Ignition Switch. Controls electrical current to forklift functions and starter motor. **(12)**

A2.4.5.9. Service Brakes/Inching Pedal, many forklifts utilize a clutch/inching pedal. During operation, this foot control disengages the transmission from the engine for the first few inches of depression of the pedal and applies the brake thereafter. This function allows the operator to maintain high engine rpm for fast operation of the load mechanism while traveling at slow speed. This function saves wear and tear on the transmission during load movement. Some forklifts have three pedals; the pedal to the far left is the inching pedal, the center pedal is the standard brake, and the right pedal is the accelerator. Other forklifts have just one inching/brake pedal combined. **(13)**

A2.4.5.10. Selector Switch. Selects Gas or LPG for Bi-Fuel engines. **(14)**

A2.4.5.11. Unusual noises. Listen for any unusual sounds, noises, erratic motions, or vibrations. Use the five senses to check for problems (look, listen, smell, feel). If encountered, the operator should stop the forklift and investigate the cause, if cause cannot be determined, then report to the Squadron VCO and/or the supervisor.

A2.4.5.12. Listen for exhaust and air leaks. Exhaust leaks are often confused with other sounds. Exhaust systems can leak from anywhere, though the mating flanges at the cylinders; silencer and tailpipe extensions are the usual spots. Another typical exhaust sound is a whistle as the engine is accelerated. For air leaks, the operator may hear the escaping of air, a continuous hissing noise. Also, low air pressure may be an indicator there may be an air leak.

Attachment 3

PERFORMANCE TEST

PERFORMANCE TEST CHECKLIST

Instructions to Certifier and Trainee:

Pre-Test Briefing: The Certifier will review this entire checklist with the trainee and brief the intended scenario that will need to be accomplished. The trainee will be given additional directions and instructions as needed as the trainee proceeds through the scenario.

Scoring Awareness: The Certifier will be scoring the trainee on forklift operations and also the general safe driving practices. The Certifier will give directions and instructions to the trainee in sufficient time for the trainee to execute. The trainee will not be asked to drive in an unsafe manner.

Conduct of Test: No Trainer or Certifier assists are permitted during the evaluation. Results will be explained to the trainee upon the conclusion of the test.

Safety: The Certifier will stop the test immediately if unsafe operations occur or health concerns arise.

SECTION I. TRAINEE AND CERTIFIER INFORMATION

1.Trainee Name (Last, First MI)	**2. Unit/Office Symbol**
3. Certifier Name (Last, First MI)	**4. Unit/Office Symbol**

SECTION II. PERFORMANCE TEST INFORMATION

5. Vehicle Management Code for Vehicle Tested	**6. Date**

SECTION III. SCORESHEET

7. Equipment & Posture	Go	No Go
Verifies vehicle required forms & personal protective equipment	☐	☐

8. Pre-Operational & General Safety:	Go	No Go
Conducts 360 degree safety inspection when approaching vehicle	☐	☐
Inspects items identified on DAF1800, documents discrepancies	☐	☐
Conducts ops check of controls & instruments	☐	☐
Signs DAF1800	☐	☐
Performs post trip inspection,reports any discrepancies to Veh Mgt	☐	☐

9. Use of Vehicle Controls:	Go	No Go
Understands all gauges, switches, levers, and buttons	☐	☐
Raises and lowers forks appropriately	☐	☐
Releases and engages parking brake appropriately	☐	☐

10. Basic Control & Operation:	Go	No Go
Avoids jerky starts and stops	☐	☐
Understands basic forklift stability of load and while traveling	☐	☐
Maintains proper speed to ensure safety	☐	☐
Tines are at proper level while traveling	☐	☐
Properly parks forklift	☐	☐

11. Knowledge of Vehicle & Use of Controls:	Go	No Go
Engine	☐	☐
Clutch and transmission	☐	☐
Brakes	☐	☐
Tires	☐	☐
Mast and tines	☐	☐

12. Cargo Loading/Unloading:	Go	No Go
Understands and follows spotter's directions	☐	☐
Ensures load is properly balanced and stacked	☐	☐
Uses tie-down straps (if applicable)	☐	☐
Safely loads/unloads cargo	☐	☐

13. General Driving Habits:	Go	No Go
Is alert and attentive	☐	☐
Adjusts to changing conditions	☐	☐
Follows all general and local safety procedures	☐	☐

14. Remarks:

SECTION IV. CERTIFIER'S ENDORSEMENT

I, the undersigned, do attest and certify that the trainee named herein has successfully completed all required training and is deemed proficient in the operation of the vehicle(s)/equipment listed in **Section II**. This certification is made to the best of my knowledge and belief, based on training conducted pursuant to the approved qualification training package.

14. Signature	**15. Date**

Figure A3.1. Performance Test Checklist.

Attachment 4

SEVEN-STEP INSPECTION PROCESS

Figure A4.1. Seven-Step Inspection Process.

SEVEN-STEP VEHICLE INSPECTION GUIDE

Step 1: Vehicle Overview

Review Last Vehicle Inspection Report. Drivers may have to make a vehicle inspection report in writing each day. The motor carrier must repair any items in the report that affect safety and certify on the report that repairs were made or were unnecessary. Only sign the previous report after verifying that any noted defects have been repaired or certified as not needing repair..

Step 2: Check Engine Compartment

Parking Brakes Are On and/or Wheels Chocked.

You may have to raise the hood, tilt the cab (ssecure any loose items to prevent them from falling and causing damage), or open the engine compartment door.

Check the following:

- Engine oil level.
- Coolant level in radiator; condition of hoses.

Check That the

- Power steering fluid level; hose condition (if so equipped).
- Windshield washer fluid level

Battery fluid level, connections, and tie downs (battery may be located elsewhere)

- Automatic transmission fluid level (may require engine to be running).
- Check belts for tightness and excessive wear (alternator, water pump, air compressor) --learn how much "give" the belts should have when adjusted right and check each one.
- Leaks in the engine compartment (fuel, coolant, oil, power steering fluid, hydraulic fluid, battery fluid).
- Cracked, worn electrical wiring insulation.
- Lower and secure hood, cab, or engine compartment door.

Step 3: Start Engine and Inspect Inside the Cab

Get In and Start Engine

Make sure parking brake is on.

Put gearshift neutral (or "park" if automatic).

Start engine; listen for unusual noises.

If equipped, check the Anti-lock Braking System (ABS) indicator lights. Light on dash should come on and then turn off. If it stays on the ABS is not working properly. For trailers only, if the yellow light on the left rear of the trailer stays on, the ABS is not working properly.

Look at the Gauges

Oil pressure. Should come up to normal within seconds after engine is started. See Figure 2.5

Air pressure. Pressure should build from 50 to 90 psi within 3 minutes. Build air pressure to governor cut-out (usually around 120 – 140 psi. **Know your vehicle's requirements.**

Ammeter and/or voltmeter. Should be in normal range(s).

Coolant temperature. Should begin gradual rise to normal operating range.

Engine oil temperature. Should begin gradual rise to normal operating range.

Warning lights and buzzers. Oil, coolant, charging circuit warning, and antilock brake system lights should go out right away.

Check Condition of Controls. Check all of the following for looseness, sticking, damage, or improper setting:

- Steering wheel.
- Clutch.
- Accelerator ("gas pedal").
- Brake controls.
- Foot brake.
- Trailer brake (if vehicle has one).
- Parking brake.
- Retarder controls (if vehicle has them).
- Transmission controls.
- Interaxle differential lock (if vehicle has one).
- Horn(s).
- Windshield wiper/washer.
- Lights.
- Headlights.
- Dimmer switch.
- Turn signal.
- Four-way flashers.
- Parking, clearance, identification, marker switch(es).

OIL PRESSURE

- Idling 5-20 PSI
- Operating 35-75 PSI
- Low, Dropping, Fluctuating: **STOP IMMEDIATELY!** Without oil the engine can be destroyed rapidly

Figure 2.5

Check Mirrors and Windshield. Inspect mirrors and windshield for cracks, dirt, illegal stickers, or other obstructions to seeing clearly. Clean and adjust as necessary.

Check Emergency Equipment

Check for safety equipment:

Spare electrical fuses (unless vehicles have circuit breakers).

Three red reflective triangles, 6 fuses or 3 liquid burning flares.

Properly charged and rated fire extinguisher.

Check for optional items such as:

- Chains (where winter conditions require).
- Tire changing equipment.
- List of emergency phone numbers
- Accident reporting kit (packet).

Check Safety Belt. Check that the safety belt is securely mounted, adjusts; latches properly and is not ripped or frayed.

Step 4: Turn Off Engine and Check Lights

Make sure the parking brake is set, turn off the engine, and take the key with you. Turn on headlights (low beams) and four-way emergency flashers and get out of the vehicle.

Step 5: Do Walk-around Inspection

- Go to front of vehicle and check that low beams are on, and both of the four-way flashers are working.
- Push dimmer switch and check that high beams work.
- Turn off headlights and four-way emergency flashers.
- Turn on parking, clearance, side-marker, and identification lights.
- Turn on right turn signal and start walk-around inspection.

General

- Walk around and inspect.
- Clean all lights, reflectors, and glass as you go along.

Left Front Side

- Driver's door glass should be clean.
- Door latches or locks should work properly.
- Left front wheel.
- Condition of wheel and rim--missing, bent, broken studs, clamps, lugs, or any signs of misalignment.
- Condition of tires--properly inflated, valve stem and cap OK, no serious cuts, bulges, or tread wear.
- Use wrench to test rust-streaked lug nuts, indicating looseness.
- Hub oil level OK, no leaks.
- Left front suspension.
- Condition of spring, spring hangers, shackles, U-bolts.
- Shock absorber condition.
- Left front brake.
- Condition of brake drum or disc.
- Condition of hoses.

Front

- Condition of front axle and steering system.
- No loose, worn, bent, damaged or missing parts.
- Must grab steering mechanism to test for looseness.
- Condition of windshield.
- Check for damage and clean if dirty.
- Check windshield wiper arms for proper spring tension.
- Check wiper blades for damage, "stiff" rubber, and securement.
- Lights and reflectors.
- Parking, clearance, and identification lights clean, operating, and proper color (amber at front).
- Reflectors clean and proper color (amber at front).
- Right front turn signal light clean, operating, and proper color (amber or white on signals facing forward).

Right Side

- Right front: check all items as done on left front.
- Primary and secondary safety cab locks engaged (if cab-over-engine design).
- Right fuel tank(s).
- Securely mounted, not damaged, or leaking.
- Fuel crossover line secure.
- Tank(s) contain enough fuel.
- Cap(s) on and secure.
- Condition of visible parts.
- Rear of engine--not leaking.
- Transmission--not leaking.
- Exhaust system--secure, not leaking, not touching wires, fuel, or air-lines.
- Frame and cross members--no bends or cracks.
- Air-lines and electrical wiring--secured against snagging, rubbing, wearing.
- Spare tire carrier or rack not damaged (if so equipped).
- Spare tire and/or wheel securely mounted.
- Spare tire and wheel adequate (proper size, properly inflated).
- Cargo securement (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Header board adequate, secure
- Side boards, stakes strong enough, free of damage, properly set in place
- Canvas or tarp properly secured to prevent tearing, billowing, or blocking of mirrors.
- If oversize, all required signs (flags, lamps, and reflectors) safely and properly mounted and all required permits in driver's possession.
- Curbside cargo compartment doors in good condition, securely closed, latched/locked and required security seals in place.

Right Rear

- Condition of wheels and rims--no missing, bent, or broken spacers, studs, clamps, or lugs.
- Condition of tires--properly inflated, valve stems and caps OK, no serious cuts, bulges, tread wear, tires not rubbing each other, and nothing stuck between them.
- Tires same type, e.g., not mixed radial and bias types.
- Tires evenly matched (same sizes).
- Wheel bearing/seals not leaking.
- Suspension.
- Condition of spring(s), spring hangers, shackles, and U-bolts.
- Axle secure.
- Powered axle(s) not leaking lube (gear oil).
- Condition of torque rod arms, bushings.
- Condition of shock absorber(s).
- If retractable axle equipped, check condition of lift mechanism. If air powered, check for leaks.
- Condition of air ride components.
- Brakes.
- Brake adjustment.
- Condition of brake drum(s) or discs.
- Condition of hoses--look for any wear due to rubbing.
- Lights and reflectors.
- Side-marker lights clean, operating, and proper color (red at rear, others amber).
- Side-marker reflectors clean and proper color (red at rear, others amber).

Rear

- Lights and reflectors.
- Rear clearance and identification lights clean, operating, and proper color (red at rear).
- Reflectors clean and proper color (red at rear).
- Taillights clean, operating, and proper color (red at rear).
- Right rear turn signal operating, and proper color (red, yellow, or amber at rear).
- License plate(s) present, clean, and secured.
- Splash guards present, not damaged, properly fastened, not dragging on ground, or rubbing tires.
- Cargo secure (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Tailboards up and properly secured.
- End gates free of damage, properly secured in stake sockets.
- Canvas or tarp (if required) properly secured to prevent tearing, billowing, or blocking of either the rearview mirrors or rear lights.
- If over-length, or over-width, make sure all signs and/or additional lights/flags are safely and properly mounted and all required permits are in driver's possession.
- Rear doors closed, latched/locked.

Left Side

- Check all items as done on right side, plus:
- Battery(ies)
- Battery box(es) securely mounted.
- Box has secure cover.
- Battery(ies) secured against movement.
- Battery(ies) not broken or leaking.
- Fluid in battery(ies) at proper level (except maintenance-free type).
- Cell caps present and securely tightened (except maintenance-free type).
- Vents in cell caps free of foreign material (except maintenance-free type).

Step 6: Check Signal Lights

- Get In and Turn Off Lights
- Turn off all lights.
- Turn on stop lights (apply trailer hand brake or have a helper put on the brake pedal).
- Turn on left turn signal lights.
- Get Out and Check Lights
- Left front turn signal light clean, operating and proper color (amber or white on signals facing the front).
- Left rear turn signal light and both stop lights clean, operating, and proper color (red, yellow, or amber).

Get In Vehicle

- Turn off lights not needed for driving.
- Check for all required papers, trip manifests, permits, etc.
- Secure all loose articles in cab (they might interfere with operation of the controls or hit you in a crash).
- Start the engine.

Step 7: Start the Engine and Check

Test for Hydraulic Leaks. If the vehicle has hydraulic brakes, pump the brake pedal three times. Then apply firm pressure to the pedal and hold for five seconds. The pedal should not move.

If it does, there may be a leak or other problem. Get it fixed before driving. If the vehicle has air brakes, do the checks described in Sections 5 and 6 of this manual.

Brake System

Test Parking Brake(s)

- Fasten safety belt
- Set parking brake (power unit only).
- Release trailer parking brake (if applicable).
- Place vehicle into a low gear.
- Gently pull forward against parking brake to make sure the parking brake holds.
- Repeat the same steps for the trailer with trailer parking brake set and power unit parking brakes released (if applicable).
- If it doesn't hold vehicle, it is faulty; get it fixed.

Test Service Brake Stopping Action

- Go about five miles per hour.
- Push brake pedal firmly
- "Pulling" to one side or the other can mean brake trouble.
- Any unusual brake pedal "feel" or delayed stopping action can mean trouble.
- If any defects are found during the inspection, they must be repaired before operating the vehicle. Federal and state laws prohibit driving an unsafe vehicle.

Attachment 5

FORKLIFT OPERATION REVIEW QUESTIONS

A5.1. Knowledge Exam Overview. This test is designed to test each student's knowledge of the forklift. Content was obtained from the Occupational Safety and Health Administration (OSHA) Forklift Operation and Safety Course Test. In order to pass the test, students must achieve at least 70% accuracy.

A5.1.1. Date Updated: April 2017.

A5.2. Forklift.

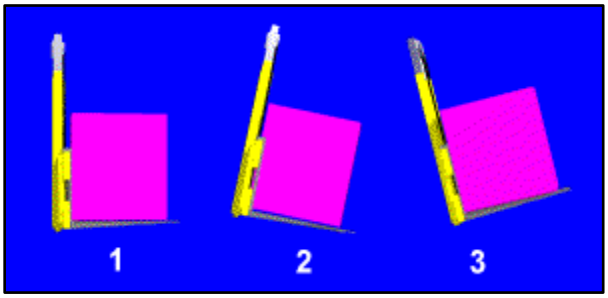
A5.2.1. When parking or leaving your truck, you should:

A. Park or leave your truck in a safe area away from traffic.
B. Lower the forks until they are flat on the floor.
C. Turn-off the engine.
D. Set the parking brake and set the directional control to neutral.
E. All of the above.

A5.2.2. All industrial trucks (lift trucks) are equipped with controls which allow you to raise/lower and tilt the forks.

A. True
B. False

A5.2.3. Of the three load positions illustrated below, which is the most stable?



A. 1
B. 2
C. 3

A5.2.4. Rear wheel steering is used on lift trucks because it gives the operator greater control when using the forks.

A. True
B. False

A5.2.5. Always check the air pressure in tires from the side, not by facing the tread.

A. True
B. False

A5.2.6. All industrial trucks (lift trucks) are equipped with a clutch.

A. True
B. False

A5.2.7. Which of the following is NOT a type of lift truck?

A. Gasoline Powered
B. Diesel Powered
C. Electric Powered
D. Air-cooled Powered

A5.2.8. Wide and long loads are more unstable than other types of loads.

A. True
B. False

A5.2.9. If the lift mechanism on your lift fails, you should try to repair the chains or hydraulic system yourself.

A. True
B. False

A5.2.10. The load capacity of a truck can be found on its data plate.

A. True
B. False

A5.2.11. It is part of your job to complete a Daily Operator's Report after you have thoroughly inspected your lift truck.

A. True
B. False

A5.2.12. If your truck starts to tip over:

A. Do not jump.
B. Stay in your seat.
C. Grip the wheel securely.
D. Brace yourself with your feet.

E. All of the above.

A5.2.13. You can stand under the forks if the engine of the lift truck is turned-off.

A. True
B. False

A5.2.14. A brake pedal that sinks to the floor under continued pressure is in good operating condition.

A. True
B. False

A5.2.15. You can drive a lift truck over any type of surface.

A. True
B. False

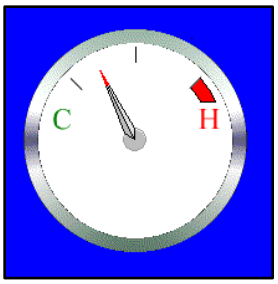
A5.2.16. The most important safety device on your lift truck is _____.

A. Horn
B. Seat Belt
C. Warning Light
D. Backup Alarm
E. You

A5.2.17. Anyone who has a valid driver's license can operate a lift truck.

A. True
B. False

A5.2.18. The gauge below is a/an:

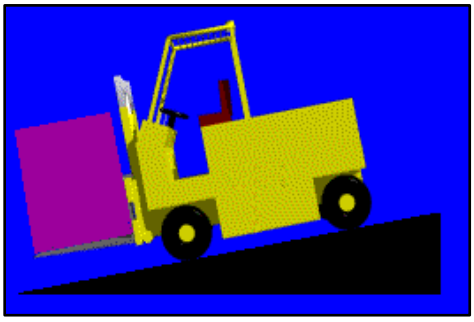


A. Gas Gauge
B. Amperes Gauge
C. Oil Gauge
D. Total Hour Meter
E. None of the above

A5.2.19. The three major parts of a lift truck are the body (truck), overhead guard, and hydraulic lift.

A. True
B. False

A5.2.20. What is wrong in the illustration below?



A. Driving in the wrong direction
B. Load is too high
C. Nothing is wrong

A5.2.21. In order to solve an "Inch Pound Equation", you must use information provided on a truck's data plate.

A. True
B. False

A5.2.22. The front wheels of a lift truck serve as the _____ between the weight of the truck and the weight of the load being carried.

A. Balance Point
B. Fulcrum Point
C. Center of Gravity
D. Seesaw Center

A5.2.23. Which of the following should NOT be allowed during the refueling or recharging process?

A. Park your lift truck in a designated refueling/recharging area.
B. Do not block doorways or access to production or emergency equipment.
C. Keep a flame burning nearby to burn off unwanted vapors or gases.
D. Check to see that there is a fire extinguisher nearby.

A5.2.24. It is safe to give someone a ride on your lift truck.

A. True
B. False

A5.2.25. Before loading or unloading a trailer at a loading dock, you should:

A. Inspect the floor of the trailer to ensure that it will support the lift truck and the load.
B. Chock the wheels of the trailer.
C. Make sure that dock plates, boards, and ramps are in place and secure.
D. All of the above.

A5.2.26. When transporting a load, you should not raise your load more than 6" from the ground.

A. True
B. False

A5.2.27. You can place your hands and feet outside of the operator's compartment, as long as your head and body are protected.

A. True
B. False

A5.2.28. Lift trucks use a hydraulic cylinder attached to chains to raise and lower the forks.

A. True
B. False

A5.2.29. How often should operators inspect their forklifts, at a minimum?

A. Hourly
B. Weekly
C. Each shift
D. Monthly

A5.2.30. When mounting and dismounting a forklift, always face_____.

A. Away from the forklift
B. Toward the forklift
C. Do it the most comfortable way for you
D. Jump on or off

A5.3. Answer Key.

5.2.1. E. All of the above
5.2.2. A. True
5.2.3. C. 3
5.2.4. A. True
5.2.5. B. False
5.2.6. B. False
5.2.7. D. Air-cooled powered
5.2.8. A. True
5.2.9. B. False
5.2.10. A. True
5.2.11. A. True
5.2.12. E. All of the above
5.2.13. B. False
5.2.14. B. False
5.2.15. B. False
5.2.16. E. You
5.2.17. B. False
5.2.18. E. None of the above
5.2.19. A. True
5.2.20. A. Driving in the wrong direction
5.2.21. A. True
5.2.22. B. Fulcrum point
5.2.23. C. Keep a flame burning nearby to burn off unwanted vapors or gases
5.2.24. B. False
5.2.25. D. All of the above
5.2.26. A. True
5.2.27. B. False
5.2.28. A. True
5.2.29. C. Each shift
5.2.30. B. Toward the forklift